

Virtual Technical Conference Platform

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Use-Case Flow

Use Case: Host Live Session

Primary Actor: Conference Author

Supporting Actors: Video Streaming & Recording Service, Track Chair, Attendee

Related Requirements: Proper A/V Readiness Check, Record Session

Preconditions: The author is authenticated and is the assigned speaker of a session that is already scheduled on the agenda. The current time is within 15 minutes of the session start.

Trigger:

The author selects “**Start Session**” from the My Sessions page.

Main Success Scenario

1. **Author opens “My Sessions” and selects the scheduled session.**
The system displays the session details, expected attendee count, and an enabled “**Start Session**” control.
2. **Author selects “Start Session”.**
The system runs **A/V Readiness Check** on the camera, microphone, speakers, and upstream bandwidth.
3. **System displays the readiness report.**
The system displays the readiness report with all four checks passed and provides a self-view preview from the camera.
4. **Author selects the presentation materials.**
The author selects the slide deck to present and chooses whether the session should be recorded using **Record Session**.
5. **Author enters the virtual room.**
The author enters the virtual room in the on-stage role. The system admits waiting attendees at the scheduled start time and displays the live attendee count.
6. **System opens the outbound stream.**
The system opens the outbound stream through the Video Streaming & Recording Service. Attendees who joined through receive the speaker's audio and video.
7. **Author shares the presentation.**
The author shares the screen or uploaded slide deck. The system renders it to all attendees within 5 seconds and pins it as the main view.
8. **Author presents the session.**
Attendees post questions through, and the Track Chair moderates them through. Approved questions appear in the speaker's Q&A panel, ordered by upvotes.

9. **System provides an end-time notification.**

Ten minutes before the scheduled end, the system notifies the author of the remaining time and surfaces the three highest-voted open questions.

10. **Author concludes the session.**

The author answers questions and then selects “**End Session**”.

11. **System closes the session.**

The system closes the stream and Q&A and stores the following session statistics:

- Peak concurrent attendance.
- Average watch time.
- Number of questions asked.
- Number of questions answered.

12. **System processes the recording.**

If recording was enabled, the system transcodes the recording and publishes it to the on-demand library within 30 minutes. The system then notifies the registered attendees of the session.

13. **Use case ends successfully.**

Alternate Flows

a. A/V Readiness Check Fails

a1. The system identifies the failing item and displays the measured value. For example: “Upstream bandwidth 0.8 Mbps, 2 Mbps required.”

The system blocks the author from entering the stage.

a2. The system offers device re-selection and a telephone dial-in option as an audio-only fallback.

a3. The author fixes the device or network issue and re-runs the readiness check. Upon successful completion, the flow resumes at Step 3.

a4. If the readiness check continues to fail five minutes after the scheduled start time, the system alerts the Track Chair. The Track Chair may:

- Play a pre-recorded backup video; or
- Reschedule the session slot.

b. Speaker Disconnects During the Session

b1. The system keeps the virtual room open and displays “**Speaker is reconnecting**” to attendees instead of ending the session.

b2. The author automatically rejoins within 15 seconds and retains the on-stage role. The author must restart screen sharing, after which the flow resumes at Step 7.

b3. If the speaker is absent for more than three minutes, the system promotes the Track Chair to host. The Track Chair may announce the delay or move the slot to open Q&A.

Postconditions

Success

- The session is marked “**Completed**”.
- Session statistics are stored.
- All attendees have left the virtual room.
- The recording, if enabled, is available on demand.

Failure

- The session is marked “**Not delivered**” with the reason for failure.
- The session slot remains on the agenda for rescheduling.
- No partial recording is published.